

Ricci Curvature Bounds and Compactness

Notes on Anderson's 1989 Paper

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1 Introduction

1.1 The variation formula

Let

$$g_t = g + th + O(t^2),$$

where h is an arbitrary symmetric 2-tensor. We use the standard variation formulas

$$\left. \frac{d}{dt} \right|_{t=0} dV_{g_t} = \frac{1}{2} \operatorname{tr}_g(h) dV_g$$

and

$$\left. \frac{d}{dt} \right|_{t=0} \operatorname{Scal}_{g_t} = -\langle \operatorname{Ric}, h \rangle + \operatorname{div} \operatorname{div} h - \Delta(\operatorname{tr} h).$$

Since M is closed, the last two divergence terms integrate to zero. Therefore, for the Hilbert–Einstein functional S , we have

$$\begin{aligned} DS_g(h) &= \int_M \left(-\langle \text{Ric}, h \rangle + \frac{1}{2} \text{Scal} \text{tr} h \right) dV \\ &= - \int_M \left\langle \text{Ric} - \frac{1}{2} \text{Scal} g, h \right\rangle dV. \end{aligned}$$

Thus an Einstein metric is a critical point of the Hilbert–Einstein functional.

We want to study compactness under bounds on the Ricci curvature rather than bounds on the sectional curvature.

1.2 Main results

Throughout these notes, compactness is understood with respect to the topology induced by $C^{k,\alpha}$ Cheeger–Gromov convergence.

Theorem 1.1. *The space of compact four-dimensional Riemannian manifolds M satisfying*

- (i) $|\text{Ric}| \leq c_1$;
- (ii) *for some $q > 4$ and $\varepsilon > 0$,*

$$\sup_{r \leq \varepsilon} r^{2(1-4/q)} \left(\int_{B(r)} |D^2 \text{Ric}|^{q/2} dV \right)^{2/q} \leq c_2(\varepsilon, q);$$

- (iii) $l_M \geq c_3$;
- (iv) $\text{Vol}(M) \geq c_4$;
- (v) $\text{diam}(M) \leq c_5$;
- (vi) $\dim H_2(M, \mathbb{R}) \leq c_6$

is $C^{1,\alpha}$ -compact in the Cheeger–Gromov topology.

Theorem 1.2. *Let $n \geq 2$.*

(I) *The space of compact n -dimensional Riemannian manifolds M satisfying*

- (i) $|\text{Ric}| \leq c_1$;
- (ii) *for some $q > n$ and $\varepsilon > 0$,*

$$\sup_{r \leq \varepsilon} r^{2(1-n/q)} \left(\int_{B(r)} |D^2 \text{Ric}|^{q/2} dV \right)^{2/q} \leq c_2(\varepsilon, q);$$

- (iii) $l_M \geq c_3$;
- (iv) $\text{Vol}(M) \geq c_4$;
- (v) $\text{diam}(M) \leq c_5$;
- (vi) $\int_M |R|^{n/2} dV \leq c_6$

is $C^{1,\alpha}$ -compact in the Cheeger–Gromov topology.

(II) If n is odd, condition (iii) may be dropped.

Remark. In dimension four, the curvature-integral bound is replaced by an upper bound on the second Betti number. This follows from the Gauss–Bonnet–Chern formula

$$\chi(M) = \int_M \text{Pf} \left(\frac{\Omega}{2\pi} \right) = \frac{1}{4\pi^2} \int_M \text{Pf}(\Omega),$$

where

$$\text{Pf}(\Omega) = \Omega_{12} \wedge \Omega_{34} - \Omega_{13} \wedge \Omega_{24} + \Omega_{14} \wedge \Omega_{23}, \quad \Omega_{ij} = \frac{1}{2} R_{ijkl} \omega^k \wedge \omega^l.$$

We compute the Pfaffian in detail, using τ for the alternating symbol:

$$\text{Pf}(\Omega) = \frac{1}{8} \tau_{ijkl} \Omega_{ij} \wedge \Omega_{kl} = \frac{1}{32} \tau_{ijkl} \tau_{abcd} R_{ijab} R_{klcd} \, dV.$$

Fix the order of i, j, k, l and consider permutations of a, b, c, d . For example, when $(a, b, c, d) = (k, l, i, j)$, we have $\tau_{ijkl} = \tau_{abcd}$, and the corresponding term is

$$\frac{1}{32} R_{ijkl}^2 \, dV,$$

which contributes $|\text{Rm}|^2$. The choices $(a, b, c, d) = (i, j, k, l)$ and $(a, b, c, d) = (i, k, j, l)$ contribute $|\text{Scal}|^2$ and $|\text{Ric}|^2$, respectively. Consequently,

$$\chi(M) = \frac{1}{32\pi^2} \int_M (|\text{Rm}|^2 - 4|\text{Ric}|^2 + |\text{Scal}|^2) \, dV.$$

Here the coefficient in the original paper is incorrect.

Since

$$\chi(M) = 2 - 2b_1(M) + b_2(M) \leq 2 + b_2(M),$$

we obtain an estimate of the form

$$b_2(M) + 2 \geq C \left(- \int_M |\text{Ric}|^2 \, dV + \int_M |\text{Rm}|^2 \, dV \right).$$

The lower Ricci-curvature bound and the diameter bound control the first integral. Hence condition (vi) in Theorem 1.2 follows, in dimension four, from condition (vi) in Theorem 1.1.

2 Preliminary Results

2.1 An estimate for the Sobolev constant

Theorem 2.1 (Croke’s isoperimetric inequality). *Let M^n be a closed Riemannian manifold satisfying*

$$\text{Ric}_M \geq (n-1)Kg, \quad D = \text{diam}(M).$$

Define

$$\Phi(M) = \inf_S \frac{\text{Area}(S)^n}{\min\{\text{Vol}(M_1), \text{Vol}(M_2)\}^{n-1}},$$

where S separates M into M_1 and M_2 . This is the quantity denoted by h_M in Anderson's paper. Set

$$s_K(r) = \begin{cases} \frac{\sin(\sqrt{K} r)}{\sqrt{K}}, & K > 0, \\ r, & K = 0, \\ \frac{\sinh(\sqrt{-K} r)}{\sqrt{-K}}, & K < 0. \end{cases}$$

Then

$$\Phi(M) \geq C_4(n) \left(\frac{\text{Vol}(M)}{\int_0^D s_K(r)^{n-1} dr} \right)^{n+1}, \quad C_4(n) = \frac{1}{4\alpha(n-1)\alpha(n)^{n-1}}.$$

Theorem 2.1 yields a lower bound for the Sobolev constant.

Lemma 2.2 (Volume growth). *Let $B_x(r)$ be a geodesic ball, let $S_x(r) = \partial B_x(r)$, and set $v(r) = \text{Vol } B_x(r)$. Whenever $v(r) < \frac{1}{2} \text{Vol}(M)$,*

$$\frac{(v'(r))^n}{v(r)^{n-1}} \geq h_M.$$

After integration,

$$\frac{v(r)}{r^n} \geq c_M.$$

Here h_M gives a lower bound for the Sobolev constant c_S of M : it is equivalent to the L^1 Sobolev constant and yields a lower bound for the L^2 Sobolev constant.

Proposition 2.3 (Sobolev inequality). *For every Lipschitz function f on M ,*

$$\|f\|_{L^{2n/(n-2)}} \leq \frac{1}{c_S} \|df\|_{L^2} + \text{Vol}(M)^{-2/n} \|f\|_{L^2}.$$

2.2 Computation of the curvature tensor

We first introduce a tensor identity. By the second Bianchi identity and the Ricci identity,

$$\Delta R = R * R + R * \text{Ric} + D^2 \text{Ric}, \quad (2.1)$$

where $D^2 \text{Ric}$ denotes a linear combination of second covariant derivatives of the Ricci tensor, and $A * B$ denotes a linear combination of contractions of $A \otimes B$.

More specifically,

$$\begin{aligned} R_{ijkl;rr} &= -R_{ijrk;lr} - R_{ijlr;kr} \\ &= -R_{rkij;lr} - R_{lr ij;kr} \\ &= -R_{rkij;rl} - R_{lr ij;rk} + R * R + R * \text{Ric} \\ &= D^2 \text{Ric} + R * R + R * \text{Ric}. \end{aligned}$$

Moreover,

$$D_r |R| = |R|^{-1} R_{ijkl} R_{ijkl;r}, \quad |DR|^2 = R_{ijkl;r}^2.$$

Thus the inequality $|D|R|^2 \leq |DR|^2$ is equivalent to

$$\sum_r \left(\sum_{i,j,k,l} R_{ijkl} R_{ijkl;r} \right)^2 \leq \sum_{i,j,k,l} R_{ijkl}^2 \sum_{i,j,k,l,r} R_{ijkl;r}^2,$$

which is the Cauchy–Schwarz inequality.

Using

$$\langle \Delta R, R \rangle + |DR|^2 = |R|\Delta|R| + |D|R|^2$$

and the inequality above, we obtain

$$\Delta|R| + c_1|D^2 \text{Ric}| + c_2|R|^2 \geq 0. \quad (2.2)$$

This elliptic inequality will be used in a Moser iteration to prove local boundedness of the curvature tensor.

2.3 Local boundedness of the curvature tensor

Lemma 2.4 (Local boundedness). *There are constants*

$$C = C(n, c_S, q, \text{Vol } B_1), \quad \varepsilon_0 = \varepsilon_0(n, c_S, q, \text{Vol } B_1),$$

where c_S is the Sobolev constant of the compact manifold M and q is fixed, such that the following holds. If $B(t)$ is a geodesic ball of radius t in M and

$$\int_{B(t)} |R|^{n/2} dV < \varepsilon_0,$$

then

$$\sup_{B(t/2)} |R| \leq C \left[\frac{1}{t^2} \int_{B(t)} |R|^{n/2} dV + t^{2\delta} \left(\int_{B(t)} |D^2 \text{Ric}|^{q/2} dV \right)^{2/q} \right], \quad \delta = 1 - \frac{n}{q}.$$

Proof. We give a new proof, because the proof in the original paper is incorrect. By rescaling the metric, we may assume $t = 1$. Choose

$$\frac{1}{2} \leq r < R \leq \frac{3}{4}, \quad M(R) = \sup_{B_R} |R|.$$

For any $x \in B_r$, set

$$l = \min \left\{ \frac{R-r}{2}, (1 + M(R))^{-1/2} \right\}.$$

Then $B_l(x) \subset B_R$ and $l^2 M(R) \leq 1$. Rescale the metric by

$$\tilde{g} = l^{-2}g, \quad |\tilde{R}| = l^2|R|, \quad |D^2 \widetilde{\text{Ric}}| = l^4 |D^2 \text{Ric}|.$$

Since all the relevant inequalities are scale invariant, we retain

$$-\tilde{\Delta}|\tilde{R}| \leq c_2|\tilde{R}|^2 + c_1|D^2 \widetilde{\text{Ric}}|.$$

Furthermore,

$$\|\widetilde{R}\|_{L^{n/2}(\widetilde{B}_1(x))} < \varepsilon_0, \quad \kappa := \|D_g^2 \widetilde{\text{Ric}}\|_{L^{q/2}(\widetilde{B}_1(x))} \leq l^{4-2n/q} \|D^2 \text{Ric}\|_{L^{q/2}(B_1)} =: l^{4-2n/q} k.$$

Set $u = |\widetilde{R}| + \kappa$. We then obtain the linear inequality

$$-\widetilde{\Delta}u \leq \left(c_2 |\widetilde{R}| + c_1 \frac{|D^2 \widetilde{\text{Ric}}|}{\kappa} \right) u.$$

If $\kappa = 0$, replace it by a positive number and let that number tend to zero at the end.

We use the classical Moser-iteration result for the linear inequality

$$\Delta u + fu \geq 0, \quad \|f\|_{L^{q/2}(\widetilde{B}_1)} \leq C,$$

to obtain the local bound

$$\sup_{\widetilde{B}_{1/2}(x)} |\widetilde{R}| \leq C \|u\|_{L^{n/2}(\widetilde{B}_1(x))} \leq C(\varepsilon_0 + \kappa).$$

Scaling back to the original metric gives

$$|R_x| \leq C(l^{-2}\varepsilon_0 + l^{2-2n/q}k)$$

for every $x \in B_r$. Thus

$$M(r) \leq C(l^{-2}\varepsilon_0 + l^{2-2n/q}k).$$

Since

$$l^{-2} \leq C((R-r)^{-2} + 1 + M(R)), \quad l \leq 1,$$

we obtain

$$M(r) \leq C\varepsilon_0 M(R) + \frac{C\varepsilon_0}{(R-r)^2} + Ck.$$

Choose $C\varepsilon_0 \leq \theta < \frac{1}{4}$ and iterate with

$$r_j = \frac{3}{4} - \frac{1}{2^{j+2}}.$$

This gives the desired estimate. □

2.4 The Cheeger–Gromov–Taylor estimate

Theorem 2.5 (Cheeger–Gromov–Taylor). *Let (M, g) be complete and let $p \in M$. Suppose that*

$$|\text{sec}| \leq kr^{-2}$$

and that the noncollapsing condition

$$\text{Vol } B_r(p) \geq \kappa r^n$$

holds. Then there is a constant $C = C(n, \kappa, k) > 0$ such that

$$\text{inj}(p) \geq Cr.$$

2.5 The Cheeger–Gromov compactness theorem

Theorem 2.6 (Pointed finite-order Cheeger–Gromov compactness). *Fix*

$$n \in \mathbb{N}, \quad k \in \mathbb{N}_0, \quad 0 < \alpha < 1.$$

Let (M_i^n, g_i, p_i) be a sequence of pointed, smooth, connected, complete Riemannian manifolds. Assume that

(i) the sequence is noncollapsed at the basepoints:

$$\text{inj}_{g_i}(p_i) \geq \iota_0 > 0;$$

(ii) for every fixed $R < \infty$ and every $0 \leq j \leq k$, there is a constant $C_j(R)$ independent of i such that

$$\sup_{B_{g_i}(p_i, R)} |\nabla_{g_i}^j \text{Rm}(g_i)|_{g_i} \leq C_j(R).$$

Then a subsequence, still indexed by i , converges to a pointed, connected, complete Riemannian manifold $(M_\infty^n, g_\infty, p_\infty)$ with $g_\infty \in C_{\text{loc}}^{k+1, \alpha}$:

$$(M_i, g_i, p_i) \longrightarrow (M_\infty, g_\infty, p_\infty)$$

in the pointed $C^{k+1, \alpha}$ Cheeger–Gromov sense.

3 Proof of Theorem 1.2

The Cheeger–Gromov compactness theorem treats the case in which $|\text{Rm}| \leq C$. Here we only have a bound on Ric , so we shall prove that the sectional curvature is also bounded.

We argue by contradiction. Suppose there is a sequence of Riemannian manifolds (M_i, g_i) such that

$$\sup_{M_i} |K_{M_i}| = R_i \longrightarrow \infty.$$

The classical approach is to rescale and pass to a limit. Suppose that the supremum is achieved at $x_i \in M_i$, and consider the rescaled metrics

$$h_i = R_i^{1/2} g_i. \tag{3.1}$$

This sequence satisfies

$$(1) \quad |\text{Ric}(h_i)| \leq c_1/R_i \rightarrow 0;$$

(2)

$$\sup_{r \leq \varepsilon R_i} r^{2(1-n/q)} \left(\int_{B(r)} |D^2 \text{Ric}(h_i)|^{q/2} dV \right)^{2/q} \longrightarrow 0;$$

$$(3) \quad |K(h_i)| \leq 1;$$

$$(4) \quad |K(h_i)(x_i)| = 1;$$

$$(5) \quad \int_{M_i} |R(h_i)|^{n/2} dV \leq c_6;$$

- (6) $l_{M_i}(h_i) \geq c_3 R_i \rightarrow \infty$;
(7) $\frac{v(r)}{r^n} = \frac{\text{Vol}_{h_i} B_x(r)}{r^n} \geq c_1$, when $v(r) \leq \frac{1}{2} \text{Vol}_{h_i} M_i$.
(8) $\text{inj}_{M_i}(x) \geq c_0$.

Conditions (1)–(6) follow from the bounds on (M_i, g_i) and the scaling properties. Condition (7) follows from Lemma 2.2 and scale invariance, while condition (8) follows from Theorem 2.5 by (3) and (7).

We would like to apply the local Cheeger–Gromov compactness theorem to the sequence (M_i, h_i) . However, a limit metric need only be $C^{1,\alpha}$, whereas we need curvature information, especially an analogue of (4), for the limit. It is therefore useful first to smooth the metrics and then to pass to a limit.

3.1 Ricci-flow smoothing and basic estimates

We use Ricci flow to smooth the metric. Short-time existence and Shi’s estimates give a Ricci-flow solution $h_i(t)$ on M_i . There is an $\varepsilon_1 > 0$, depending only on the dimension, such that $h_i(t)$ exists for $0 \leq t \leq \varepsilon_1$ and

$$\begin{aligned} \|h_i(t) - h_i\|_{C^0} &\leq C(t), & C(t) &\rightarrow 0 \quad \text{as } t \rightarrow 0, \\ \|D^k R_i(t)\|_{C^0} &\leq C(t^{-1}, k), & k &= 1, 2, 3, \dots, \end{aligned} \tag{3.2}$$

and

$$\|R_i(t)\|_{C^0} \leq C.$$

In the remainder of this subsection we omit the index i , since all estimates are uniform for sufficiently large i .

Lemma 3.1 (Curvature concentration). *For the metric $h = h(0)$, there are fixed positive constants r_0 and δ_0 such that*

$$\int_{B_{x_0}(r_0)} |R|^2 dV \geq \delta_0.$$

Proof. Since $|R| \leq 1$, choose r sufficiently small to apply Lemma 2.4 and obtain

$$\sup_{B_{x_0}(r/2)} |R|^2 \leq C \left[\frac{1}{r^2} \int_{B_{x_0}(r)} |R|^{n/2} dV + r^{2\delta} \left(\int_{B_{x_0}(r)} |D^2 \text{Ric}|^{q/2} dV \right)^{2/q} \right], \quad \delta = 1 - \frac{n}{q}.$$

By condition (2), the second term converges to zero as $i \rightarrow \infty$. On the other hand, condition (4) gives

$$\sup_{B_{x_0}(r/2)} |R|^2 = 1,$$

and hence

$$\int_{B_{x_0}(r)} |R|^{n/2} dV \geq cr^2.$$

The result follows once more from $|R| \leq 1$. □

Lemma 3.2. *For the metric $h = h(0)$, there is a constant K such that*

$$\int_{B_x(1)} |\nabla R|^2 dV \leq K$$

for every $x \in M$.

Proof. Equation (2.2) implies

$$\Delta|R|^2 \geq -c_1|D^2 \text{Ric}||R| - c_2|R|^3 + |\nabla R|^2.$$

Let $\zeta = \zeta(r)$ be a fixed cutoff function such that $\zeta \equiv 1$ for $r \leq 1$ and $\zeta \equiv 0$ for $r \geq 2$. Multiplying by ζ and integrating by parts gives

$$\int_{B_x(1)} |\nabla R|^2 \leq \int_M \zeta \Delta|R|^2 + c_1 \int_M \zeta |D^2 \text{Ric}||R| + c_2 \int_M \zeta |R|^3.$$

For the first term on the right-hand side,

$$\int_M \zeta \Delta|R|^2 = \int_M |R|^2 \Delta \zeta \leq \int_M |\Delta \zeta| \leq C.$$

For the second,

$$\int_M \zeta |D^2 \text{Ric}||R| \leq \int_{B_2} |D^2 \text{Ric}| \leq C$$

by condition (2). For the third,

$$\int_M \zeta |R|^3 \leq \text{Vol } B_2 \leq C$$

by the Bishop–Gromov theorem. This is the only point at which condition (2) is used. $\square$

Lemma 3.3. *There is a constant L such that, for $0 \leq t \leq \varepsilon_1$,*

$$\int_{B_x(1)} |\nabla R(t)|^2 dV_t \leq L e^{Lt}.$$

Proof. Step I: an evolution equation. We first derive the evolution formula for $|\nabla R(t)|^2$ under the Ricci flow. Since

$$\partial_t R = \Delta R + R * R,$$

taking a covariant derivative gives

$$\nabla \partial_t R = \nabla \Delta R + \nabla R * R.$$

Using

$$\nabla \partial_t A - \partial_t \nabla A = A * \nabla R, \quad \nabla \Delta A - \Delta \nabla A = \nabla A * R + A * \nabla R,$$

we obtain

$$\partial_t \nabla R = \Delta \nabla R + \nabla R * R.$$

Consequently,

$$\partial_t |\nabla R|^2 = \langle \Delta \nabla R, \nabla R \rangle + R * \nabla R * \nabla R.$$

The extra term comes from differentiating the metric $g(t)$. Since

$$\Delta \langle \nabla R, \nabla R \rangle = \langle \Delta \nabla R, \nabla R \rangle + |\nabla^2 R|^2 \geq \langle \Delta \nabla R, \nabla R \rangle,$$

we have

$$\partial_t |\nabla R|^2 \leq \Delta |\nabla R|^2 + C|R| |\nabla R|^2. \quad (3.3)$$

Step II: a maximal argument. Fix $r_1 < \frac{1}{2} \text{inj}_M$ and choose a maximal collection of $r_1/2$ -separated points $\{p_j\}$. Thus the balls $B_{r_1/4}(p_j)$ are disjoint, and the balls $B_{r_1/2}(p_j)$ cover M . By compactness, the collection is finite.

Choose a cutoff function ζ_j such that $\zeta_j \equiv 1$ on $B_{r_1/2}(p_j)$ and $\zeta_j \equiv 0$ outside $B_{r_1}(p_j)$. Define

$$E_j(t) = \int_{B_{r_1}(p_j)} |\nabla R(t)|^2 dV_t, \quad F_j(t) = \int_{B_{r_1}(p_j)} \zeta_j |\nabla R(t)|^2 dV_t,$$

and

$$F(t) = \max_j F_j(t).$$

The ball $B_{r_1}(p_j)$ can be covered by finitely many balls $B_{r_1/2}(p_{j_k})$, with $\{p_{j_k}\} \subset \{p_j\}$. The number N_j of such balls has a uniform upper bound N_0 by the Ricci-curvature bound and the Bishop–Gromov theorem. Therefore,

$$\begin{aligned} E_j(t) &\leq \sum_{j_k} \int_{B_{r_1/2}(p_{j_k})} |\nabla R(t)|^2 dV_t \\ &\leq \sum_{j_k} \int_{B_{r_1}(p_{j_k})} \zeta_{j_k} |\nabla R(t)|^2 dV_t \\ &\leq N_0 F(t). \end{aligned}$$

By (3.3),

$$\begin{aligned} F'_j(t) &= \int_{B_{r_1}(p_j)} \zeta_j |\nabla R|^2 |\text{Scal}| dV_t + \int_{B_{r_1}(p_j)} \zeta_j (\Delta |\nabla R|^2 + C |\nabla R|^2 |R|) dV_t \\ &\leq \int_{B_{r_1}(p_j)} \zeta_j |\nabla R|^2 dV_t + \int_{B_{r_1}(p_j)} |\nabla R|^2 |\Delta \zeta_j| dV_t \\ &\leq C E_j(t). \end{aligned}$$

Combining these inequalities gives

$$F'_j(t) \leq C N_0 F(t) = C F(t).$$

Since $F'(t) \leq \max_j F'_j(t)$ in the generalized sense, we have $F'(t) \leq C F(t)$. Gronwall's inequality then gives

$$F(t) \leq F(0) e^{Ct} \leq K e^{Ct}.$$

Finally, $B_x(1)$ can be covered by finitely many balls $B_{r_1}(p_j)$, with the total number uniformly controlled in i . Hence

$$\int_{B_x(1)} |\nabla R(t)|^2 dV_t \leq L e^{Lt}$$

for some L . □

Lemma 3.4. *For every $t \leq \varepsilon_1$, there is a constant C such that*

- (i) $\int_{B_{x_0}(1)} |R(t)|^2 dV_t \geq C^{-1}$;
- (ii) $\int_M |R(t)|^{n/2} dV_t \leq C$;
- (iii) $|\text{Ric}(t)| \leq C |\text{Ric}(h(0))|$.

Proof. The curvature evolution equation is

$$\partial_t |R|^2 = \Delta |R|^2 - 2|\nabla R|^2 + R * R * R.$$

For (i), as in the preceding lemma,

$$\partial_t |R|^2 \geq \Delta |R|^2 - 2|\nabla R|^2 - C|R|^3.$$

Choose a cutoff function and integrate by parts to obtain

$$\frac{d}{dt} \int_{B_x(1)} \zeta |R|^2 dV \geq - \int_{B_x(1)} |\Delta \zeta| - 2 \int_{B_x(1)} \zeta |\nabla R|^2 - C \int_{B_x(1)} \zeta |R|^3.$$

If

$$G(t) = \int_{B_x(1)} \zeta |R|^2 dV,$$

then, using $|R| \leq 1$,

$$G'(t) \geq -C - CG(t).$$

Gronwall's inequality gives $G(t) \geq C_0$ for $t \in (0, \varepsilon_0)$.

For (ii), the evolution inequality

$$\partial_t |R|^{n/2} \leq \Delta |R|^{n/2} + C|R|^{(n+2)/2}$$

and integration by parts give

$$\frac{d}{dt} \int_M |R|^{n/2} dV \leq C \int_M |R|^{n/2} dV.$$

Gronwall's inequality yields the result.

For (iii), the evolution equation

$$\partial_t \text{Ric} = \Delta \text{Ric} + R * \text{Ric}$$

implies

$$\frac{d}{dt} |\text{Ric}|^2 \leq \Delta |\text{Ric}|^2 + C|\text{Ric}|^2.$$

Let $u = e^{-Ct} |\text{Ric}|^2$. The maximum principle gives the desired estimate. $\square$

3.2 The structure of the limit space

Returning to the sequence $\{h_i\}$ on M_i , the preceding results give metrics $h_i(t)$, for $t \leq \varepsilon_1$, such that

- (i) $\|h_i(t) - h_i\|_{C^0} \leq C(t)$;
- (ii) $\|\nabla^k R_i(t)\|_{C^0} \leq C(t^{-1}, k)$;
- (iii) $\|R_i(t)\|_{C^0} \leq 1$;
- (iv) $\int_{M_i} |R_i(t)|^{n/2} dV_t \leq C$;
- (v) $\int_{B_{x_i}(1)} |R_i(t)|^2 dV_t \geq C^{-1}$;

$$(vi) \text{ Vol}_{h_i(t)} B(r) r^{-n} \geq C^{-1};$$

$$(vii) \text{ inj}_{h_i(t)} M_i \geq C^{-1}.$$

We now apply the Cheeger–Gromov compactness theorem to the pointed manifolds $(M_i, x_i, h_i(t))$. For every fixed R and $t > 0$, the sequence $(B_{x_i}(R), h_i(t))$ has a subsequence converging in the $C^{k,\alpha}$ topology, for every k , to a smooth Riemannian manifold $(B_{x_\infty}(R), h_\infty(t))$. Choosing $R_j \rightarrow \infty$ and taking a diagonal subsequence, we obtain smooth convergence on compact subsets to a smooth Riemannian manifold N with metric $h_N(t)$, $0 < t < \varepsilon_1$, satisfying

$$(i) \text{ Ric}_N = 0;$$

$$(ii) \int_N |R|^{n/2} dV < \infty;$$

$$(iii) v(r)/r^n \geq C;$$

$$(iv) \int_{B_z(1)} |R|^2 dV > 0$$

for some $z \in N$.

Theorem 3.5 (Structure at infinity). *Let N be a complete, noncompact Riemannian manifold satisfying conditions (i)–(iii) above. Given $z \in N$, there is an $R_0 > 0$ such that*

$$N \setminus B_z(R_0) \cong (R_0, \infty) \times S^{n-1}(1)/\Gamma,$$

where $S^{n-1}(1)/\Gamma$ is a spherical space form. Moreover, N is asymptotically flat: its metric approaches the flat metric on $(R_0, \infty) \times S^{n-1}(1)/\Gamma$ at rate $O(r^{-2})$. If N is simply connected at infinity, so that $\Gamma = \{e\}$, then N is isometric to $\mathbb{R}^n$.

The proof proceeds by showing from conditions (i)–(iii) that N is asymptotically locally Euclidean. The lower bound on the shortest geodesic loop will then imply that it is flat, contradicting condition (iv).

Lemma 3.6 (Croke–Yau). *Let (M, g) be a Riemannian manifold, not necessarily compact, with $\text{Ric} \geq 0$. Suppose that*

$$\frac{v(r)}{r^n} \geq c_0$$

for every $r > 0$. Then the global Sobolev inequality

$$\|f\|_{L^{2n/(n-2)}} \leq c_S \|Df\|_{L^2}$$

holds.

By Lemma 3.6, N has a global Sobolev constant.

Let $d = \dim N$. Fix $n \in \mathbb{N}$ and consider a component

$$A_n^r := A(r/n, nr) = \left\{ x \in N : \frac{r}{n} < d(z, x) < nr \right\}$$

of the annulus. Since

$$\int_{A_n^r} |R|^{d/2} dV \longrightarrow 0 \quad \text{as } r \rightarrow \infty,$$

Lemma 2.4 gives, for sufficiently large $r > R_0$,

$$\sup_{S(r)} |R| \leq \frac{c}{r^2} \int_{A_n^r} |R|^{d/2} dV = \frac{\varepsilon(r)}{r^2}, \quad \varepsilon(r) \rightarrow 0 \quad \text{as } r \rightarrow \infty.$$

Set $g^r = r^{-2}g$. Then the rescaled annuli (A_n^r, g^r) satisfy

$$\sup_{A_n^r} |R^r| \leq n^2 \varepsilon(r) \rightarrow 0.$$

The diameter of A_n^r is bounded by $2n$. Since the volume is noncollapsed and the sectional curvature is bounded, the Cheeger–Gromov–Taylor estimate gives

$$\text{inj}_{g^r}(x) \geq \delta_0 > 0$$

for every $x \in A_n^r$. The Cheeger–Gromov compactness theorem therefore gives a flat limit A_n^∞ for every n .

By a diagonal argument, choose radii r_i such that $A(r_i/n, nr_i)$ converges to A_n^∞ for every n , and

$$A_n^\infty \subset A_m^\infty \quad \text{if } m \geq n.$$

Letting $n \rightarrow \infty$, we obtain a flat Riemannian manifold B_∞ . The compatibility among the different A_n^∞ is described as follows. There are homeomorphisms

$$F_i^m : A_m^\infty \rightarrow A_m^{r_i}$$

and isometric embeddings

$$\iota_{m,n} : A_m^\infty \rightarrow A_n^\infty, \quad m < n,$$

such that

$$\iota_{m,n} = \lim_{i \rightarrow \infty} (F_i^n)^{-1} \circ F_i^m.$$

Define

$$B_\infty := \bigsqcup_i A_i^\infty / \sim, \quad (x, m) \sim (\iota_{m,n}(x), n).$$

The distance functions ρ_i on $A_n^{r_i}$ induce a function on B_∞ by

$$\rho = \lim_{i \rightarrow \infty} \rho_i \circ (F_i^n)^{-1}.$$

Claim 1. $|\nabla \rho| = 1$ almost everywhere.

Proof. The function ρ is the limit of Lipschitz functions whose Lipschitz constants tend to 1, so $|\nabla \rho| \leq 1$. Conversely, let x be a differentiability point of ρ . Choose $x_i \in A_n^{r_i}$ with $\rho_i(x_i) \rightarrow \rho(x)$. If γ_i is a minimizing geodesic from x_i to z , then

$$\rho_i(x_i) = \rho_i(\gamma_i(s)) + s.$$

By the Arzelà–Ascoli theorem, a subsequence converges to a curve γ in B_∞ , and

$$\rho(x) = \rho(\gamma(s)) + s.$$

Hence $|\nabla \rho| \geq |d\rho(\gamma')| = 1$. □

Claim 2. Define $C(s) = \rho^{-1}(s)$. Then

$$d(C(s), C(t)) = |s - t|.$$

Proof. The inequality $d(C(s), C(t)) \geq |s - t|$ is immediate. For the reverse inequality, we prove that

$$d(y, C(s)) \leq |s - t|$$

for $y \in C(t)$ and $s < t$. Choose $y_i \in A_n^{r_i}$ such that $\rho_i(y_i) \rightarrow \rho(y) = t$. Let γ_i be a minimizing geodesic from z to y_i and set $x_i = \gamma_i(s)$. By the Arzelà–Ascoli theorem, a subsequence of γ_i converges to a geodesic γ in B_∞ , with $x_i \rightarrow x = \gamma(s)$. Therefore,

$$d(y, C(s)) \leq d(y, x) = \lim_{i \rightarrow \infty} d_i(y_i, x_i) = |s - t|.$$

□

Claim 3. We have

$$\lim_{s \rightarrow 0} \text{diam } C(s) = 0.$$

Proof. Consider

$$D(s/2) = \{x \in B_\infty : d(x, C(s)) \leq s/2\}.$$

Then $\rho(D(s/2)) \leq 3s/2$. Pulling back to $A_n^{r_i}$ and using the Bishop–Gromov theorem gives

$$\text{Vol } D(s/2) \leq c_1 s^d.$$

Choose a covering of $D(s/2)$ such that the balls $\{B_{s/8}(x_l)\}$ are disjoint and the balls $\{B_{s/4}(x_l)\}$ cover $D(s/2)$. Pulling back once more gives

$$\text{Vol } B_{s/4}(x_l) \geq c_2 s^d.$$

If $N_s = \#\{x_l\}$, then

$$N_s c_2 s^d \leq c_1 s^d,$$

so $N_s \leq C$, independently of s . Any two points in $C(s)$ can be joined by a chain of balls from $\{B_{s/4}(x_l)\}$; hence

$$\text{diam } C(s) \leq \frac{N_s s}{4} \rightarrow 0 \quad \text{as } s \rightarrow 0.$$

□

The three claims allow us to attach a point to B_∞ and define

$$\overline{B}_\infty = B_\infty \sqcup \{0\},$$

extending ρ to $\overline{B}_\infty$. This is a complete, proper metric space and therefore, by the Hopf–Rinow theorem, a geodesic length space. In particular, any two points can be joined by a minimizing geodesic. Denote a minimizing geodesic joining 0 to $x \in C(1)$ by γ_x .

Gap. The argument above does not establish that the sets $C(s)$ coincide with metric spheres. We therefore skip this point and assume that, for all sufficiently large r , the annulus $A(r/n, nr)$ is diffeomorphic to

$$(r/n, nr) \times S^{d-1}/\Gamma.$$

Equivalently, we assume that, for some $R_0 > 0$,

$$A(R_0, \infty) = (R_0, \infty) \times S^{d-1}/\Gamma.$$

Under this assumption, every component of $N \setminus B_{R_0}$ has this form. We claim that $N \setminus B_{R_0}$ is connected. Otherwise, N has at least two ends, and the Arzelà–Ascoli theorem yields a line in N . The Cheeger–Gromoll splitting theorem gives

$$N \cong N_1 \times \mathbb{R}, \quad \text{Ric}_{N_1} = 0.$$

There are three possibilities.

1. If N_1 is noncompact and has more than two ends at infinity, the same argument gives another splitting, so $N \cong N_2 \times \mathbb{R}^2$. This is impossible because $N \setminus B_{R_0}$ is disconnected.
2. If N_1 is noncompact and has one end at infinity, this is also impossible because N_1 is connected at infinity whereas $N \setminus B_{R_0}$ is not.
3. If N_1 is compact, then $\text{AVR}(N) = 0$, contradicting $\text{AVR}(N) > 0$.

It remains to consider the case $\Gamma = \{e\}$. Then

$$\begin{aligned} \lim_{r \rightarrow \infty} \frac{\text{Vol } B_r(z)}{r^d} &\geq \lim_{r \rightarrow \infty} \frac{\text{Vol } A(r/m, r)}{r^d} \\ &= \lim_{r \rightarrow \infty} \text{Vol } A^r(1/m, 1) \\ &= \text{Vol } B_1 - \text{Vol } B_{1/m}. \end{aligned}$$

Since m is arbitrary,

$$\lim_{r \rightarrow \infty} \frac{\text{Vol } B_r(z)}{r^d} \geq \text{Vol } B_1.$$

The rigidity statement in the Bishop–Gromov theorem implies that N is flat.

3.3 Completion of the proof

We now use the length of the shortest geodesic loop to prove that $\Gamma = \{e\}$. Recall that the rescaled sequence satisfies

$$(1) \quad |\text{Ric}(h_i)| \leq c_1/R_i \rightarrow 0;$$

$$(2)$$

$$\sup_{r \leq \varepsilon R_i} r^{2(1-n/q)} \left(\int_{B(r)} |D^2 \text{Ric}(h_i)|^{q/2} dV \right)^{2/q} \rightarrow 0;$$

$$(3) \quad |K(h_i)| \leq 1;$$

$$(4) \quad |K(h_i)(x_i)| = 1;$$

$$(5) \quad \int_{M_i} |R(h_i)|^{n/2} dV \leq c_6;$$

$$(6) \quad l_{M_i}(h_i) \geq c_3 R_i \rightarrow \infty;$$

$$(7) \quad \frac{v(r)}{r^n} = \frac{\text{Vol}_{h_i} B_x(r)}{r^n} \geq c_1;$$

$$(8) \quad \text{inj}_{M_i}(x) \geq c_0.$$

We focus on condition (6). The preceding argument describes the structure of the limit space N : there is an $R_0 > 0$ such that

$$N \setminus B_{R_0} \cong [R_0, \infty) \times S^{n-1}/\Gamma.$$

3.3.1 The homotopy groups

We prove that every homotopy group of N vanishes. First consider $\pi_k(N)$ for $k \geq 2$. By the spectral sequence, it is equivalent to determine $\pi_k(\Omega_z N)$ for $k \geq 1$.

Let

$$f : S^k \longrightarrow \Omega_z N$$

be continuous and map the basepoint of S^k to the constant path in $\Omega_z N$. Since S^k is connected, $f(s)$ is a null-homotopic path in N . As S^k is compact, the energy $E(f(s))$ is bounded above by a constant A .

Let $U = B_z(R)$ be a sufficiently large domain in N , to be fixed below. For sufficiently large i , pull U back to M_i as

$$M_i \supset U_i = F_i(U).$$

If i is large enough, the length l_i of the shortest null-homotopic geodesic loop satisfies $l_i \geq \sqrt{2A}$. The image of $f(s)$ in M_i therefore has energy below that of the shortest geodesic loop.

Define the cutoff loop space

$$\Omega_{x_i, \leq B}, \quad B = \frac{l_i^2}{2} + 1.$$

The map f induces

$$\tilde{f} : S^k \longrightarrow \Omega_{x_i, \leq B}, \quad \tilde{f}(s) = (t \mapsto F_i(f(s)(t))).$$

A classical result in Morse theory gives a homotopy between $\Omega_{x_i, \leq B}$ and the constant path during which the energy does not increase. If U is chosen sufficiently large, the homotopy remains in U , because its energy is below B . We can therefore move the entire homotopy back to $\Omega_z N$, proving that $[f] = 0$.

It remains to treat $\pi_1(N)$. A classical result of Li and Anderson states that a manifold with $\text{AVR} > 0$ has finite fundamental group. Consider the universal cover $\tilde{N}$. By the Whitehead theorem, $\tilde{N}$ is contractible because all its homotopy groups vanish. Thus

$$\chi(\tilde{N}) = 1.$$

Since $\chi(N)$ divides 1, the only possibility is $N = \tilde{N}$; hence N is simply connected.

The Lefschetz–Poincaré duality theorem and the Mayer–Vietoris sequence show that $\partial B_z(R_0)$ is a homology sphere. In particular,

$$H_1(\partial B_z(R_0)) = 0 = \Gamma_{\text{ab}}.$$

If $|\Gamma|$ is sufficiently small, for example $|\Gamma| \leq 5$, then necessarily $\Gamma = \{e\}$. In fact, we shall prove that $|\Gamma| \leq 3$.

3.3.2 The order of Γ

Assume that $|\Gamma| \geq 4$. Pull the canonical radial coordinate on the truncated cone back to $N \setminus B_{R_0}$ and denote it by ρ . Thus

$$\partial(N \setminus B_{R_0}) = \rho^{-1}(R_0).$$

As $R \rightarrow \infty$, the hypersurface $\rho^{-1}(R)$ approaches S^{n-1}/Γ , so for sufficiently large R the hypersurface $\rho^{-1}(R)$ is convex.

Define

$$D(R) = B_{R_0} \cup \rho^{-1}([R_0, R]).$$

Using the $C^{1,\alpha}$ Cheeger–Gromov convergence $M_i \rightarrow N$, pull these domains back to (M_i, g_i) . Let

$$B_i = F_i(D(R)), \quad B_{0,i} = F_i(D(R_0)).$$

For sufficiently large i ,

$$\text{cl}(B_i \setminus B_{0,i}) \cong [R_0, R] \times S^{n-1}/\Gamma,$$

and ∂B_i is strictly convex.

Fix $p \in \partial D(R)$ and set $p_i = F_i(p)$. Let γ be the shortest loop based at p_i that is not null-homotopic in $B_i \setminus B_{0,i}$. We claim that, when R is sufficiently large,

$$\gamma \subseteq B_i \setminus B_{0,i}.$$

It follows that γ is a geodesic loop. The loop cannot leave B_i because B_i is convex. If it meets the inner boundary $\partial B_{0,i}$, then, for sufficiently large i ,

$$L(\gamma) \geq 2(1 - \varepsilon)(R - R_0), \tag{3.4}$$

because B_i approaches the corresponding geodesic ball in M_i .

On the other hand, we construct a loop based at p_i with length at most θR for some $\theta < 2$. Let

$$m = |\Gamma| \geq 4$$

and choose a lift $\tilde{p} \in S^{n-1}(R)$ of $p \in S^{n-1}(R)/\Gamma$. After dividing the radius by R , regard it as a point of the unit sphere. Since Γ acts freely, the orbit

$$\{g\tilde{p} : g \in \Gamma\}$$

consists of m distinct points, denoted by $x_1, \dots, x_m$. Since

$$0 \leq \left| \sum_{a=1}^m x_a \right|^2 = m + 2 \sum_{a < b} \langle x_a, x_b \rangle,$$

there is at least one pair x_a, x_b such that

$$\langle x_a, x_b \rangle \geq -\frac{1}{m-1}.$$

Therefore,

$$d_{S^{n-1}(1)}(x_a, x_b) \leq \arccos \left(-\frac{1}{m-1} \right) \leq \arccos \left(-\frac{1}{3} \right).$$

Set

$$\theta = \arccos\left(-\frac{1}{3}\right) < 2.$$

The corresponding path projects to a nontrivial loop in $S^{n-1}(R)/\Gamma$. Hence, for sufficiently large i , there is a loop σ based at p_i with

$$L(\sigma) \leq (1 + \varepsilon)\theta R. \quad (3.5)$$

Combining (3.4) and (3.5), we conclude that

$$\gamma \Subset B_i \setminus B_{0,i}.$$

Thus γ is a geodesic loop with length less than $(1 + \varepsilon)\theta R$. This upper bound is independent of i , contradicting

$$l_i \longrightarrow \infty \quad \text{as } i \rightarrow \infty.$$

Therefore $|\Gamma| \leq 3$, so $\Gamma = \{e\}$. The limit manifold is flat, contradicting the curvature concentration in Lemma 3.1. This completes the proof of Theorem 1.2.